File

Name: P4-fit-A2
Description: KISSsoft example
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1 Overview

Calculation method: According to DIN 7190-1:2017, elastic range
with additions such as centrifugal force, micro sliding, mounting, etc.
for a horizontal or longitudinal interference fit

Diameter shaft (mm)	[DiA]/[DiI]	50.00	/ 0.00			
Diameter hub (mm)	[DaA]/[DaI]	100.00	/50.00			
Length of Interference fit (mm)	[l]	40.00				
Diameter of joint (mm)	[DF]	50.00				
with special allowances:						
Upper allowance Shaft (µm)	[AoI]	75.0				
Lower allowance Shaft (µm)	[AuI]	75.0				
Tolerance measure Shaft (µm)	[TI]	0.0				
Upper allowance Hub (µm)	[AoA]	0.0				
Lower allowance Hub (µm)	[AuA]	0.0				
Tolerance measure Hub (µm)	[TA]	0.0				
Largest interference (µm)	[Po]	75.0				
Smallest interference (µm)	[Pu]	75.0				
Nominal torque (Nm)	[T]	0.00				
Application factor	[KA]	1.00				
Service torque (Nm)	[Tb]	0.00				
Axial force (N)	[FA]	0.00				
Bending moment (Nm)	[Mb]	0.00				
Radial force (N)	[Fr]	0.00				
Circumferential force (N)	[Fu]	0.00				
Speed (1/min)	[n]	0.00				
Interference (µm)	[P]	75.0	(75.0	..	75.0	)
Embedding (µm)	[s]	3.84				
Effective interference at 20°C(68°F) (µm)	[Pw]	71.16	(71.2	..	71.2	)
Effective interference at working temp.. (µm)	[PwTh]	71.16	(71.2	..	71.2	)
Service temperature shaft (°C)	[ThB]	20.0				
Service temperature hub (°C)	[ThB]	20.0				
Pressure stress by						
- Interference (after mounting) (N/mm²)	[pM]	114.75	(114.75	..	114.75	)
- Interference (working) (N/mm²)	[p]	114.75	(114.75	..	114.75	)
- Bending moment (N/mm²)	[pb]	0.00				
- Radial force (N/mm²)	[pr]	0.00				
Coefficient of friction, longitudinal	[µa]	0.120				
Coefficient of friction, circumferential	[µu]	0.120				
Safety against sliding	[Sr]	> 1000				
Required safety against sliding	[SSr]	1.50				

Equivalent stress according to von Mises
$\sigma V = (\sigma R^2+\tau T^2-\sigma R \cdot \tau T)^{0.5}$

1.1 Shaft

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[El]	215000.00			
Poisson's ratio (-)	[vl]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (component's raw diameter) (N/mm²)	[Rm]	650.00			
Yield point (component's raw diameter) (N/mm²)	[Rp]	200.00			
Component raw diameter (mm)	[deff]	55			
Surface class of roughness	N6 Rz=4.8 (Grinding)				
Surface roughness (µm)	[Rz]	4.80			
External diameter (mm)	[DiA]	50.00			
Diameter increase during operation (µm)	[ΔD]	-18.68	(-18.68 ..	-18.68)	
Diameter increase during operation (o/oo)	[ΔD]	-0.37	(-0.37 ..	-0.37)	
Equivalent stress outer diameter (N/mm²)	[σVa]	114.75	(114.75 ..	114.75)	
- Radial stress (N/mm²)	[σRa]	-114.75	(-114.75 ..	-114.75)	
- Tangential stress (N/mm²)	[σTa]	-114.75	(-114.75 ..	-114.75)	
- With outside load (N/mm²)	[σVaMF]	114.75	(114.75 ..	114.75)	
Inner diameter (mm)	[Dil]	0.00			
Equivalent stress inner diameter (N/mm²)	[σVi]	114.75	(114.75 ..	114.75)	
- Radial stress (N/mm²)	[σRi]	-114.75	(-114.75 ..	-114.75)	
- Tangential stress (N/mm²)	[σTi]	-114.75	(-114.75 ..	-114.75)	
Safety against fracture	[SiRm]	5.66	(5.66 ..	5.66)	
Required safety against fracture	[SSi.Rm]	1.50			
Safety against yield point	[Si.Rp]	1.74	(1.74 ..	1.74)	
Required safety against yield point	[SSi.Rp]	1.10			

1.2 Hub

Material, own Input		C45 (1)			
Young's modulus (N/mm²)	[EA]	215000.00			
Poisson's ratio (-)	[vA]	0.30			
Density (kg/m³)	[ρ]	7830.00			
Coefficient of thermal expansion (10^-6/K)	[α]	11.50			
Tensile strength (wall thickness) (N/mm²)	[Rm]	700.00			
Yield point (wall thickness) (N/mm²)	[Rp]	300.00			
The hub material strength values are defined using the wall thickness as the raw diameter.					
Component raw diameter (wall thickness) (mm)	[deff]	26			
Surface class of roughness	N6 Rz=4.8 (Grinding)				
Surface roughness (µm)	[Rz]	4.80			
External diameter (mm)	[DaA]	100.00			
Diameter increase during operation (µm)	[ΔD]	35.58	(35.58 ..	35.58)	
Diameter increase during operation (o/oo)	[ΔD]	0.36	(0.36 ..	0.36)	
Equivalent stress outer diameter (N/mm²)	[σVa]	76.50	(76.50 ..	76.50)	
- Radial stress (N/mm²)	[σRa]	0.00	(0.00 ..	0.00)	
- Tangential stress (N/mm²)	[σTa]	76.50	(76.50 ..	76.50)	
Inner diameter (mm)	[DaI]	50.00			
Diameter increase during operation (µm)	[ΔD]	52.48	(52.48 ..	52.48)	
Diameter increase during operation (o/oo)	[ΔD]	1.05	(1.05 ..	1.05)	
Equivalent stress inner diameter (N/mm²)	[σVi]	267.74	(267.74 ..	267.74)	
- Radial stress (N/mm²)	[σRi]	-114.75	(-114.75 ..	-114.75)	
- Tangential stress (N/mm²)	[σTi]	191.24	(191.24 ..	191.24)	
- Radial stress with outside load (N/mm²)	[σRai]	-114.75	(-114.75 ..	-114.75)	
- Tangential stress with outside load (N/mm²)	[σTai]	191.24	(191.24 ..	191.24)	
- With outside load (N/mm²)	[σViMF]	267.74	(267.74 ..	267.74)	

Safety against fracture	[Si.Rm]	2.61	(	2.61	..	2.61)
Required safety against fracture	[SSi.Rm]	1.50				
Safety against yield point	[Si.Re]	1.12	(	1.12	..	1.12)
Required safety against yield point	[SSi.Re]	1.10				

Mere elastic stress, no verification of elastic plastic interference fit according to DIN 7190-1.

2 Service / Mounting / Dismounting

Transverse-interference-fit:						
Mounting clearance (mm)	[PsTh]	0.050				
Temperature difference for mounting:						
Shaft temperature: (°C) Hub temperature: [ThA] (°C)						
Mount with shaft temperature 20 °C:						
Assembly temperature shaft (°C)	[θI]	20				
Assembly temperature hub (°C)	[θA]	237.391				
Mount with shaft temperature -150 °C:						
Assembly temperature shaft (°C)	[θI]	-150				
Assembly temperature hub (°C)	[θA]	111.739				
calculated using the coefficient of thermal expansion						
Sub-cooling of the shaft according to DIN 7190-1 (10^-6/K)	[αw]	8.50				
Longitudinal pressure fit:						
Assembly temperature shaft (°C)	[θM]	20.00				
Assembly temperature hub (°C)	[θM]	20.00				
Coefficient of friction, longitudinal	[μe=μa*	1.300]	0.16			
Press on (force) (kN)	[Fpress]	112.47	(112.47	..	112.47)	
Coefficient of friction, longitudinal	[μll=μa*1.600]	0.19				
Press out (force) (kN)	[Fpress]	138.43	(138.43	..	138.43)	
Coefficient of friction	[μ]	0.19				
Max. torque to avoid Micro sliding (Nm)	[Tlimit]	662.21	(662.21	..	662.21)	

Notice concerning the display: Number-1 (Number-2.. Number-3):
- Number-1: Value calculated with the mean allowance
- Number-2: Value of the smallest possible allowance
- Number-3: Value of the largest possible allowance

Notice: All strains and stresses are calculated for the purely elastic case.

End of report (lines: 154)